

# Static Holographic Boundary Theory

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## Abstract

Static Holographic Boundary Theory (SHBT) addresses the problem of deriving a single consistency principle that links Standard Model branch data, gravitational closure, finite observation, and the baryon asymmetry without introducing independent phenomenological knobs. The key insight is that the primitive object should be a completed boundary conformal field theory, not a pre-existing bulk spacetime: exact modular invariance of the completed torus partition function supplies the closure condition from which the bulk and observer sectors are rendered. The residual failure of this closure is measured by a framing defect  $\Delta_{\text{fr}}$ .

The paper develops the canonical branch  $(k_\ell, k_q, K) = (26, 8, 312)$ , for which the integer locks  $I_\ell^* = 6$  and  $I_q^* = 13$  give  $\Delta_{\text{fr}} = 0$ . On this branch, modular closure is equivalent to a vanishing bulk closure tensor and to the projected Hamiltonian constraint  $\Pi_{\text{phys}} H_{\text{total}} \Pi_{\text{phys}} = 0$ . Finite boundary entropy densities are then mapped through a holographic renormalization-group pipeline into positive, trace-normalized metric slices; anti-baryon de-rendering produces the benchmark visible asymmetry  $\eta_b = 6.449923359416 \times 10^{-10}$  while preserving passive stress-energy; and Causal Points crystallize only histories that fit within a local entropy budget.

The significance of SHBT is therefore its boundary-first closure program: the same modular data determine visible-sector arithmetic, dark completion, effective geometry, baryogenesis, and finite observation, with each claim paired to an auditable implementation check.

## 1 Introduction

Quantum gravity and Standard Model phenomenology are usually formulated in opposite directions. Perturbative quantum field theory starts from local particle multiplets, gauge charges, and renormalization-group data, and then asks how this successful low-energy structure survives coupling to gravity. Holography starts from boundary data and asks how local bulk spacetime and effective fields emerge. A realistic theory must make these two directions meet: it must account for the observed particle content, vacuum response, baryon asymmetry, and cosmological scales without repeatedly inserting small numbers by hand. This is the naturalness problem in its broadest form. The Standard Model is phenomenologically successful, but the gravitational and vacuum sectors appear highly sensitive to ultraviolet completion unless a deeper closure principle fixes the allowed data [1, 2, 3].

Static Holographic Boundary Theory (SHBT) proposes that the missing closure principle is exact modular invariance of a completed boundary conformal field theory (CFT). The starting point is

neither a fluctuating classical spacetime nor a pre-existing cosmological history, but a completed boundary Hilbert space and a physical torus partition function

$$Z_{\partial}(\tau, \bar{\tau}) = \sum_{I,J} M_{IJ} \chi_I(\tau) \overline{\chi_J(\tau)}, \quad [M, S_{\partial}] = [M, T_{\partial}] = 0, \quad M_{IJ} \in \mathbb{Z}_{\geq 0}. \quad (1)$$

The role of modular covariance follows the standard lesson that torus partition functions in two-dimensional CFT are not arbitrary sums of characters: their operator content is constrained by the modular group [4, 5]. SHBT strengthens this consistency condition into a first principle. A boundary sector is physical only if it closes under the completed modular kernels, and any residual mismatch is encoded in a scalar framing defect  $\Delta_{\text{fr}}$ . The central claim is the closure chain

$$[M, S_{\partial}] = [M, T_{\partial}] = 0 \iff \Delta_{\text{fr}} = 0 \iff \mathcal{E}_{\mu\nu} = 0 \iff \Pi_{\text{phys}} H_{\text{total}} \Pi_{\text{phys}} = 0. \quad (2)$$

Thus holographic closure is not imposed after a bulk theory has been guessed. Rather, the vanishing of the bulk closure tensor and the disappearance of an independent external time generator are the bulk and Hamiltonian images of completed modular closure on the boundary. The implementation audits this link through `shbt_simulator.ShbtSimulator().run_full_audit().boundary_report`, with the partition data and visible modular phases constructed by `shbt_simulator.StaticBoundary.evaluate_z_boundary`, `shbt_simulator.StaticBoundary.build_su2_visible_phases`, and `shbt_simulator.StaticBoundary.build_su3_visible_phases`.

The canonical branch studied here is

$$\mathfrak{b}_* = (k_{\ell}, k_q, K) = (26, 8, 312), \quad I_{\ell}^* = \frac{K}{2k_{\ell}} = 6, \quad I_q^* = \frac{K}{3k_q} = 13. \quad (3)$$

Within SHBT this branch is the anomaly-free fixed point because the lepton, color, and parent completion levels lock to integers and give

$$\Delta_{\text{fr}}(312, 26, 8) = \max\left(\left\|\frac{312}{2 \cdot 26}\right\|_{\mathbb{Z}}, \left\|\frac{312}{3 \cdot 8}\right\|_{\mathbb{Z}}\right) = 0. \quad (4)$$

The visible factors are  $SU(2)_{26}$  and  $SU(3)_8$ , embedded in an  $SO(10)_{312}$  parent completion and a dark completion sector. The dark sector is therefore not an optional phenomenological reservoir. It is the finite residue required by exact closure, with

$$c_{\text{dark}} = \frac{834433}{362670} \simeq 2.300805139658643, \quad 8\pi G_N = \frac{12}{c_{\text{dark}} M_P^2}. \quad (5)$$

The same arithmetic appears in `shbt_simulator.StaticBoundary.benchmark_branch`, `shbt_simulator.StaticBoundary.lepton_level`, `shbt_simulator.StaticBoundary.quark_level`, `shbt_simulator.StaticBoundary.parent_level`, `shbt_simulator.StaticBoundary.i_l_star`, `shbt_simulator.StaticBoundary.i_q_star`, `shbt_simulator.StaticBoundary.c_dark`, and `report.framing_defect`.

The construction has four linked components. First, the completed boundary CFT is treated as the primary object,

$$\mathcal{H}_{\partial}^{\text{comp}} = \mathcal{H}_{SU(2)_{26}} \otimes \mathcal{H}_{SU(3)_8} \otimes \mathcal{H}_{SO(10)_{312}} \otimes \mathcal{H}_{\text{dark}}, \quad (6)$$

whose physical spectrum is the modular-invariant, nonnegative-integer pairing of completed characters. This is the algebraic source of gauge and gravitational consistency. Second, entropy self-resolution turns the finite boundary register

$$N = N_{\text{sat}} = \frac{3\pi}{L_P^2 \Lambda_{\text{holo}}}, \quad \Lambda_{\text{holo}} > 0, \quad (7)$$

into normalized loading and entanglement densities. Their entropy gradients define a holographic renormalization-group flow into bulk metric slices. This continues the broader holographic and black-hole thermodynamic tradition [6, 7, 8], but makes the entropy budget finite and auditable through `shbt_simulator.HolographicProjection`, `shbt_simulator.BulkMetricSlice`, and the `shbt_simulator.ShbtSimulator().run_full_audit().projection_report` record.

Third, SHBT reframes baryogenesis as anti-baryon de-rendering. The observed baryon asymmetry is usually framed as a dynamical departure from baryon number, C, CP, and equilibrium conditions [9]. Here the active render charge is separated from passive stress-energy. Anti-baryon visible charges are projected out of the actively rendered gauge channels, while the passive gravitational source is preserved in the dark completion sector. The resulting asymmetry is therefore an observer-visible render imbalance, not a failure of completed stress-energy accounting. The implementation checks this claim through

- `shbt_simulator.ShbtSimulator().run_full_audit().baryogenesis_identity`,
- `shbt_simulator.BaryogenesisOptimizer.derender_antibaryon_charges`,
- `report.stress_energy_preserved`, and
- `shbt_simulator.ShbtSimulator().run_full_audit().benchmark_delta`.

The benchmark value reported by the audit is `eta_b=6.449923359416e-10`.

Fourth, the Causal Point makes the observer part of the same finite register. SHBT does not assume an external classical observer with unlimited memory. An observer is a finite local address equipped with a light-cone sample, a local property packet, a hidden-state budget, and a history projection operator  $\Pi_{A,\iota}$ . A measurement is admitted only when the residual entropy budget is nonnegative. Observable time and crystallized history are therefore finite-register projections rather than primitive background parameters. The corresponding code objects are `shbt_simulator.CausalPoint`, `shbt_simulator.LightConeSample`, `shbt_simulator.LocalPropertyPacket`, `shbt_simulator.CoordinateLogEntry`, the `shbt_simulator.ShbtSimulator().run_full_audit().memory_report` record, and `shbt_simulator.CausalPoint.crystallize_history`.

The main results follow from the same closure data. Because the branch (26, 8, 312) has exact framing closure, the bulk closure tensor vanishes and the topological Einstein equation follows in the SHBT sector:

$$\mathcal{E}_{\mu\nu} = \frac{c_{\text{dark}} M_P^2}{12} (G_{\mu\nu} + \Lambda_{\text{holo}} g_{\mu\nu}) - T_{\mu\nu}^{\text{vis}} = 0. \quad (8)$$

The static physical sector has no independent external time generator because the completed Hamiltonian constraint annihilates physical states,  $\Pi_{\text{phys}} H_{\text{total}} \Pi_{\text{phys}} = 0$ . Finite-register entropy flow produces symmetric, trace-normalized, positive metric slices and an observer-frame effective cosmology.

The Causal Point construction turns measurement into an entropy-budgeted projection with explicit failure conditions. Anti-baryon de-rendering yields the benchmark baryon asymmetry while preserving passive stress-energy. Running `python examples/run_audit.py` is the intended reproducibility check: it should report the canonical branch, `framing_defect=0.0`, `modular_invariant=True`, `zero_energy_locked=True`, the finite bit budget, nine bulk metric slices, nine crystallized history entries, `eta_b=6.449923359416e-10`, and `stress_energy_preserved=True`.

This positioning clarifies the relationship between SHBT and existing approaches. Like AdS/CFT, SHBT treats a boundary quantum theory as sufficient data for a bulk gravitational description [10, 11, 12]. Unlike standard AdS/CFT, the present construction is not organized around a large- $N$  conformal gauge theory with an asymptotically anti-de Sitter bulk. It is organized around a completed static boundary CFT, modular closure, and a finite holographic register. Like entropic-gravity programs, SHBT treats geometry as downstream of information and entropy [13]; unlike a pure entropic-force derivation, its entropy gradients are constrained by modular closure and checked slice-by-slice by an implementation audit. Like topological and constraint-based approaches to quantum gravity, SHBT removes an external time generator from the physical sector; unlike a purely formal constraint algebra, it ties that removal to visible-sector branch arithmetic, dark completion, baryogenesis, and observer memory.

The paper is organized as follows. Section 2 derives the canonical branch and states the axiomatic closure conditions. Section 3 constructs the modular data and boundary partition function. Section 4 defines the finite entropy densities that self-resolve the boundary register, and Section 5 maps their gradients into bulk metric slices. Section 6 develops anti-baryon de-rendering and stress-energy preservation, while Section 7 defines Causal Points and history crystallization. Section 8 collects the implementation audits, and Section 9 summarizes the unified closure picture and remaining limitations. Throughout, the mathematical derivation and `shbt_simulator.ShbtSimulator().run_full_audit()` are kept in parallel so that the same branch, defect, entropy budget, metric projection, Causal Point, and baryogenesis identity must close both on paper and in code.

## 2 Canonical Branch and Axiomatic Foundation

The canonical branch is derived by treating the completed boundary theory, not a bulk Lagrangian, as the primitive object. For arbitrary visible levels  $(k_\ell, k_q)$  and parent level  $K$ , the completed boundary Hilbert space is postulated to factor as

$$\mathcal{H}_\partial^{\text{comp}}(k_\ell, k_q, K) = \mathcal{H}_{SU(2)_{k_\ell}} \otimes \mathcal{H}_{SU(3)_{k_q}} \otimes \mathcal{H}_{SO(10)_K} \otimes \mathcal{H}_{\text{dark}}. \quad (9)$$

The collective primary index is written  $I = (i_\ell, i_q, i_{10}, \alpha)$ , and its completed character is

$$\chi_I(\tau) = \chi_{i_\ell}^{(k_\ell)}(\tau) \chi_{i_q}^{(k_q)}(\tau) \chi_{i_{10}}^{(K)}(\tau) \chi_\alpha^{\text{dark}}(\tau). \quad (10)$$

The physical torus partition function has the form

$$Z_\partial(\tau, \bar{\tau}) = \sum_{I, J} M_{IJ} \chi_I(\tau) \overline{\chi_J(\tau)}, \quad M_{IJ} \in \mathbb{Z}_{\geq 0}, \quad (11)$$

with nonnegative integer multiplicities. The first axiom is that the completed pairing matrix is invariant under the full boundary modular kernels,

$$[M, S_\partial] = 0, \quad [M, T_\partial] = 0, \quad (12)$$

where

$$\begin{aligned} S_\partial &= S_{SU(2)_{k_\ell}} \otimes S_{SU(3)_{k_q}} \otimes S_{SO(10)_K} \otimes S_{\text{dark}}, \\ T_\partial &= T_{SU(2)_{k_\ell}} \otimes T_{SU(3)_{k_q}} \otimes T_{SO(10)_K} \otimes T_{\text{dark}}. \end{aligned} \quad (13)$$

Equivalently,  $Z_\partial(\tau + 1, \bar{\tau} + 1) = Z_\partial(\tau, \bar{\tau})$  and  $Z_\partial(-1/\tau, -1/\bar{\tau}) = Z_\partial(\tau, \bar{\tau})$ , so the allowed operator content is a modular-invariant completed pairing rather than an arbitrary list of boundary primaries [4, 5].

The integral-spin condition is the local form of the  $T$ -constraint. A character transforms as

$$\chi_I(\tau + 1) = \exp\left[2\pi i \left(h_I - \frac{c_I}{24}\right)\right] \chi_I(\tau), \quad (14)$$

while the anti-holomorphic character contributes the conjugate phase. Therefore the summand with coefficient  $M_{IJ}$  is  $T$ -invariant only if

$$M_{IJ} \neq 0 \implies \exp\left\{2\pi i \left[h_I - h_J - \frac{c_I - c_J}{24}\right]\right\} = 1, \quad (15)$$

or, equivalently,

$$\boxed{M_{IJ} \neq 0 \implies h_I - h_J - \frac{c_I - c_J}{24} \in \mathbb{Z}.} \quad (16)$$

This condition is the boundary algebraic precursor of the SHBT framing defect. The visible conformal weights used by the implementation are

$$\begin{aligned} h_a^{SU(2)_k} &= \frac{a(a+2)}{4(k+2)}, \\ h_{(p,q)}^{SU(3)_k} &= \frac{p^2 + q^2 + pq + 3p + 3q}{3(k+3)}, \end{aligned} \quad (17)$$

with central charges

$$c_{SU(2)_k} = \frac{3k}{k+2}, \quad c_{SU(3)_k} = \frac{8k}{k+3}, \quad c_{SO(10)_K} = \frac{45K}{K+8}. \quad (18)$$

The canonical branch is the minimal parent completion compatible with the visible lepton and quark levels used throughout the audit. The embedding locks require the parent level to close simultaneously over the lepton and color descendants,

$$I_\ell = \frac{K}{2k_\ell} \in \mathbb{Z}, \quad I_q = \frac{K}{3k_q} \in \mathbb{Z}. \quad (19)$$

For the visible levels  $k_\ell = 26$  and  $k_q = 8$ , the smallest positive parent level satisfying both locks is

$$K = \text{lcm}(2 \cdot 26, 3 \cdot 8) = \text{lcm}(52, 24) = 312. \quad (20)$$

Thus

$$\mathbf{b}_* = (k_\ell, k_q, K) = (26, 8, 312), \quad I_\ell^* = \frac{312}{2 \cdot 26} = 6, \quad I_q^* = \frac{312}{3 \cdot 8} = 13. \quad (21)$$

This is the mathematical counterpart of `shbt_simulator.StaticBoundary.benchmark_branch = (26, 8, 312)`, together with `shbt_simulator.StaticBoundary.lepton_level`, `shbt_simulator.StaticBoundary.quark_level`, `shbt_simulator.StaticBoundary.parent_level`, `shbt_simulator.StaticBoundary.i_l_star`, and `shbt_simulator.StaticBoundary.i_q_star`.

The scalar obstruction to these integer locks is the framing defect

$$\Delta_{\text{fr}}(K, k_\ell, k_q) = \max\left(\left\|\frac{K}{2k_\ell}\right\|_{\mathbb{Z}}, \left\|\frac{K}{3k_q}\right\|_{\mathbb{Z}}\right), \quad \|x\|_{\mathbb{Z}} = \min_{n \in \mathbb{Z}} |x - n|. \quad (22)$$

At the benchmark branch both arguments are integers, so

$$\Delta_{\text{fr}}(312, 26, 8) = \max(\|6\|_{\mathbb{Z}}, \|13\|_{\mathbb{Z}}) = 0. \quad (23)$$

The branch is therefore anomaly-free in the SHBT sense: the parent completion has no residual visible framing mismatch. In code this is reported as `report.framing_defect=0.0` and checked by the `shbt_simulator.ShbtSimulator().run_full_audit().boundary_report` record.

The second axiom identifies the bulk closure tensor as the covariant image of the scalar modular obstruction. Define

$$\mathcal{E}_{\mu\nu} \equiv \frac{c_{\text{dark}} M_P^2}{12} (G_{\mu\nu} + \Lambda_{\text{holo}} g_{\mu\nu}) - T_{\mu\nu}^{\text{vis}}, \quad 8\pi G_N = \frac{12}{c_{\text{dark}} M_P^2}. \quad (24)$$

SHBT postulates a nondegenerate closure map

$$\mathcal{E}_{\mu\nu} = \mathcal{Q}_{\text{fr}} \Delta_{\text{fr}} \mathcal{P}_{\mu\nu}, \quad \mathcal{Q}_{\text{fr}} \neq 0, \quad \mathcal{P}_{\mu\nu} \neq 0, \quad (25)$$

so vanishing of the scalar obstruction is equivalent to vanishing of its bulk image:

$$\Delta_{\text{fr}} = 0 \iff \mathcal{E}_{\mu\nu} = 0. \quad (26)$$

Using Eq. (24), the condition  $\mathcal{E}_{\mu\nu} = 0$  is precisely

$$G_{\mu\nu} + \Lambda_{\text{holo}} g_{\mu\nu} = \frac{12}{c_{\text{dark}} M_P^2} T_{\mu\nu}^{\text{vis}} = 8\pi G_N T_{\mu\nu}^{\text{vis}}. \quad (27)$$

Hence the canonical SHBT closure chain is

$$\Delta_{\text{fr}} = 0 \iff \mathcal{E}_{\mu\nu} = 0 \iff G_{\mu\nu} + \Lambda_{\text{holo}} g_{\mu\nu} = 8\pi G_N T_{\mu\nu}^{\text{vis}} \quad (28)$$

The tensor equation is therefore not an independent tuning condition; it is the bulk representative of exact modular closure.

The final axiom is static global Hamiltonian cancellation. Let

$$H_{\text{total}} = H_{\partial} + H_{\text{bulk}} + H_{\text{stabilizer}}, \quad (29)$$

where  $H_\partial$  acts on the completed boundary characters,  $H_{\text{bulk}}$  is the bulk image of the closed boundary data, and  $H_{\text{stabilizer}}$  enforces the closure syndromes. The physical projector is defined by the simultaneous constraints

$$\Pi_{\text{phys}}M = M\Pi_{\text{phys}} = M, \quad P_{\text{fr}}\Pi_{\text{phys}} = 0, \quad \mathcal{E}_{\mu\nu}\Pi_{\text{phys}} = 0. \quad (30)$$

On this sector the stabilizer generator is fixed, not freely chosen:

$$H_{\text{stabilizer}}\Pi_{\text{phys}} = -(H_\partial + H_{\text{bulk}})\Pi_{\text{phys}}. \quad (31)$$

Multiplying Eq. (29) by  $\Pi_{\text{phys}}$  on both sides gives

$$\Pi_{\text{phys}}H_{\text{total}}\Pi_{\text{phys}} = 0. \quad (32)$$

Therefore, for every physical completed-boundary state satisfying  $|\Psi_{\text{phys}}\rangle = \Pi_{\text{phys}}|\Psi_{\text{phys}}\rangle$ ,

$$\boxed{H_{\text{total}}|\Psi_{\text{phys}}\rangle = 0.} \quad (33)$$

This is the static-block statement that the completed physical sector has no independent external time generator.

The Rust/Python simulator mirrors these axioms through the `shbt_simulator.StaticBoundary` class. The constructor fixes the branch data; the visible modular phases are assembled by `shbt_simulator.StaticBoundary.build_su2_visible_phases` and `shbt_simulator.StaticBoundary.build_su3_visible_phases`; the finite partition check is performed by `shbt_simulator.StaticBoundary.evaluate_z_boundary`; the scalar obstruction is computed by `report.framing_defect`; and the closure chain, including  $\Delta_{\text{fr}} = 0$ , the Einstein lock, and global Hamiltonian cancellation, is audited by `shbt_simulator.ShbtSimulator().run_full_audit().boundary_report`.

### 3 Modular Data and Boundary Partition Function

This section makes explicit the modular data used by the completed boundary partition function. The general Wess–Zumino–Witten rule is that an affine factor  $G_k$  has integrable highest weights  $\lambda$ , modular character vector  $\chi^{G_k}$ , dual Coxeter number  $g^\vee$ , central charge

$$c_{G_k} = \frac{k \dim G}{k + g^\vee}, \quad (34)$$

and diagonal modular- $T$  entries

$$T_{\lambda\mu}^{G_k} = \delta_{\lambda\mu} \exp\left[2\pi i \left(h_\lambda^{G_k} - \frac{c_{G_k}}{24}\right)\right], \quad h_\lambda^{G_k} = \frac{(\lambda, \lambda + 2\rho)}{2(k + g^\vee)}. \quad (35)$$

Here  $\rho$  is the Weyl vector and the inner product is normalized so long roots have length squared 2. The Rust/Python simulator stores the reduced visible phase  $\exp(2\pi i h_\lambda)$  in `t_boundary`; the omitted scalar factor  $\exp(-2\pi i c/24)$  is common to a fixed factor and cancels between the holomorphic and anti-holomorphic sectors, and it also leaves the commutator norms unchanged.

For  $SU(2)_k$ , the integrable labels are  $a = 0, 1, \dots, k$ . With spin  $j = a/2$  and  $C_2(j) = j(j+1) = a(a+2)/4$ , Eq. (35) gives

$$h_a^{SU(2)_k} = \frac{a(a+2)}{4(k+2)}, \quad c_{SU(2)_k} = \frac{3k}{k+2}. \quad (36)$$

The  $A_1$  Weyl sum has two terms and reduces to the sine kernel

$$S_{ab}^{SU(2)_k} = \sqrt{\frac{2}{k+2}} \sin\left(\frac{\pi(a+1)(b+1)}{k+2}\right), \quad T_{ab}^{SU(2)_k} = \delta_{ab} \exp\left[2\pi i \left(h_a^{SU(2)_k} - \frac{c_{SU(2)_k}}{24}\right)\right]. \quad (37)$$

For  $SU(3)_k$ , the integrable labels are Dynkin pairs  $(p, q)$  with  $p, q \geq 0$  and  $p+q \leq k$ . The quadratic Casimir is

$$C_2(p, q) = \frac{p^2 + q^2 + pq + 3p + 3q}{3}, \quad (38)$$

so that

$$h_{(p,q)}^{SU(3)_k} = \frac{p^2 + q^2 + pq + 3p + 3q}{3(k+3)}, \quad c_{SU(3)_k} = \frac{8k}{k+3}. \quad (39)$$

It is useful to embed a Dynkin label in the traceless three-dimensional weight space by

$$v(p, q) = \left(\frac{2p+q}{3}, \frac{-p+q}{3}, \frac{-p+2q}{3}\right), \quad \rho = v(1, 1) = (1, 0, -1). \quad (40)$$

Then the  $A_2$  Weyl-sum formula used by the implementation is

$$S_{(p,q),(r,s)}^{SU(3)_k} = \frac{i^3}{\sqrt{3}(k+3)} \sum_{\sigma \in S_3} \text{sgn}(\sigma) \exp\left[-\frac{2\pi i}{k+3}(\sigma(v(p, q) + \rho), v(r, s) + \rho)\right], \quad (41)$$

with

$$T_{(p,q),(r,s)}^{SU(3)_k} = \delta_{(p,q),(r,s)} \exp\left[2\pi i \left(h_{(p,q)}^{SU(3)_k} - \frac{c_{SU(3)_k}}{24}\right)\right]. \quad (42)$$

For the parent completion  $SO(10)_K$ , the Lie algebra is  $D_5$ , with  $g^\vee = 8$ , rank 5,  $\dim SO(10) = 45$ , and 20 positive roots. Thus

$$c_{SO(10)_K} = \frac{45K}{K+8}. \quad (43)$$

If  $\Lambda$  and  $M$  are integrable  $D_5$  highest weights, the corresponding formal kernels are

$$S_{\Lambda M}^{SO(10)_K} = \frac{i^{20}}{2(K+8)^{5/2}} \sum_{w \in W(D_5)} \text{sgn}(w) \exp\left[-\frac{2\pi i}{K+8}(w(\Lambda + \rho), M + \rho)\right],$$

$$T_{\Lambda M}^{SO(10)_K} = \delta_{\Lambda M} \exp\left[2\pi i \left(\frac{(\Lambda, \Lambda + 2\rho)}{2(K+8)} - \frac{45K}{24(K+8)}\right)\right]. \quad (44)$$

The finite audit does not enumerate the full  $SO(10)_{312}$  character table; instead, this parent factor enters through the branch locks, the framing-defect test, and the central charge in Eq. (43).

At the benchmark branch, the visible block evaluated in code is the  $3 \times 3$  slice

$$\text{charge\_embedding} = (k_\ell - 4, k_\ell - 3, k_\ell) = (22, 23, 26), \quad (45)$$

$$\text{LOW\_SU3\_WEIGHTS} = ((0, 0), (1, 0), (0, 1)). \quad (46)$$

This slice is not the full completed Hilbert space. It is the visible finite-dimensional block on which `shbt_simulator.StaticBoundary.s_boundary`, `shbt_simulator.StaticBoundary.t_boundary`, and `shbt_simulator.StaticBoundary.evaluate_z_boundary` perform the numerical audit. The combined visible index is

$$A = (i, j), \quad a_i \in (22, 23, 26), \quad w_j \in ((0, 0), (1, 0), (0, 1)), \quad (47)$$

so the visible audit space has 9 basis entries. In that ordering,

$$S_{\partial}^{\text{code}} = \text{Re } S_{SU(2)_{26}}^{\text{vis}} \otimes \text{Re } S_{SU(3)_8}^{\text{vis}}, \quad T_{\partial}^{\text{code}} = \text{diag}\left(T_{SU(2)_{26}}^{\text{vis}} \otimes T_{SU(3)_8}^{\text{vis}}\right), \quad (48)$$

which is exactly the construction of the `s_boundary` and `t_boundary` properties.

The benchmark central charges and conformal weights are

$$\begin{aligned} c_{SU(2)_{26}} &= \frac{39}{14} = 2.785714285714, & c_{SU(3)_8} &= \frac{64}{11} = 5.818181818182, \\ c_{SO(10)_{312}} &= \frac{351}{8} = 43.875, & c_{\text{vis}} &= \frac{1325}{154} = 8.603896103896, \\ h_{22}^{SU(2)_{26}} &= \frac{33}{7} = 4.714285714286, & h_{23}^{SU(2)_{26}} &= \frac{575}{112} = 5.13928571429, \\ h_{26}^{SU(2)_{26}} &= \frac{13}{2} = 6.5, & h_{(0,0)}^{SU(3)_8} &= 0, \\ h_{(1,0)}^{SU(3)_8} &= \frac{4}{33} = 0.121212121212, & h_{(0,1)}^{SU(3)_8} &= \frac{4}{33} = 0.121212121212. \end{aligned} \quad (49)$$

The visible  $S$  blocks produced by `shbt_simulator.StaticBoundary.build_su2_visible_block` and `shbt_simulator.StaticBoundary.build_su3_visible_block` are

$$\begin{aligned} S_{SU(2)_{26}}^{\text{vis}} &\simeq \begin{pmatrix} 0.088270792276 & -0.208953252971 & 0.142191553507 \\ -0.208953252971 & 0.260560444585 & -0.115960306962 \\ 0.142191553507 & -0.115960306962 & 0.029923764933 \end{pmatrix}, \\ \text{Re } S_{SU(3)_8}^{\text{vis}} &\simeq \begin{pmatrix} 0.018018539201 & 0.048334858719 & 0.048334858719 \\ 0.048334858719 & 0.110479936109 & 0.110479936109 \\ 0.048334858719 & 0.110479936109 & 0.110479936109 \end{pmatrix}. \end{aligned} \quad (51)$$

The imaginary parts of the displayed  $SU(3)$  entries are at roundoff scale for the first row and are conjugate in the fundamental/anti-fundamental subblock; the implemented `s_boundary` uses the real part shown above. The reduced visible  $T$  phases used by the code are

$$\begin{aligned} T_{SU(2)_{26}}^{\text{vis}} &\simeq \begin{pmatrix} -0.222520933956 - 0.974927912182i \\ 0.666346577952 + 0.745642164883i \\ -1.000000000000 \end{pmatrix}, \\ T_{SU(3)_8}^{\text{vis}} &\simeq \begin{pmatrix} 1 \\ 0.723734038105 + 0.690079011482i \\ 0.723734038105 + 0.690079011482i \end{pmatrix}. \end{aligned} \quad (52)$$

The completed boundary partition function is the torus pairing of completed characters,

$$Z_{\partial}(\tau, \bar{\tau}) = \sum_{I, J} M_{IJ} \chi_I(\tau) \overline{\chi_J(\tau)}, \quad M_{IJ} \in \mathbb{Z}_{\geq 0}, \quad (53)$$

where  $I = (a, (p, q), \Lambda, \alpha)$  includes the  $SU(2)$  label,  $SU(3)$  label,  $SO(10)$  parent label, and dark-sector label. In vector notation this is  $Z_\partial = \chi^\dagger M \chi$ , up to the conventional placement of the holomorphic index. Since the character vector transforms as

$$\chi(\tau + 1) = T_\partial \chi(\tau), \quad \chi(-1/\tau) = S_\partial \chi(\tau), \quad (54)$$

modular invariance follows from

$$[M, T_\partial] = 0, \quad [M, S_\partial] = 0. \quad (55)$$

Indeed,

$$Z_\partial(\tau + 1, \bar{\tau} + 1) = \chi(\tau)^\dagger T_\partial^\dagger M T_\partial \chi(\tau) = \chi(\tau)^\dagger M \chi(\tau), \quad (56)$$

$$Z_\partial(-1/\tau, -1/\bar{\tau}) = \chi(\tau)^\dagger S_\partial^\dagger M S_\partial \chi(\tau) = \chi(\tau)^\dagger M \chi(\tau), \quad (57)$$

using unitarity of the modular kernels. Therefore

$$\boxed{Z_\partial(\tau + 1, \bar{\tau} + 1) = Z_\partial(\tau, \bar{\tau}), \quad Z_\partial(-1/\tau, -1/\bar{\tau}) = Z_\partial(\tau, \bar{\tau}).} \quad (58)$$

The finite Rust/Python partition audit takes  $M_{\text{code}} = \mathbb{K}_9$  and the leading visible character monomials

$$\chi_{(i,j)}^{\text{code}}(\tau) = q^{h_{a_i}^{SU(2)26} + h_{w_j}^{SU(3)8} - c_{\text{vis}}/24}, \quad q = e^{2\pi i \tau}, \quad c_{\text{vis}} = \frac{1325}{154}. \quad (59)$$

Thus `shbt_simulator.StaticBoundary.evaluate_z_boundary` evaluates

$$Z_\partial^{\text{code}}(\tau) = \overline{\chi^{\text{code}}(\tau)}^T M_{\text{code}} \chi^{\text{code}}(\tau). \quad (60)$$

For  $\tau = i$  the code gives

$$Z_\partial^{\text{code}}(i) = 2.441381789163 \times 10^{-24},$$

$$Z_\partial^{\text{code}}(i + 1) = 2.441381789163 \times 10^{-24}, \quad (61)$$

$$Z_\partial^{\text{code}}(-1/i) = 2.441381789163 \times 10^{-24}. \quad (62)$$

The equality under the  $S$  move in this numerical line is pointwise because  $i$  is the self-dual point of  $\tau \mapsto -1/\tau$ ; the nontrivial finite-block audit of the two modular generators is the matrix commutator test. The `shbt_simulator.ShbtSimulator().run_full_audit().boundary_report` record reports

| Audit quantity                         | Value          |
|----------------------------------------|----------------|
| <code>s_boundary.shape</code>          | (9, 9)         |
| <code>t_boundary.shape</code>          | (9, 9)         |
| <code>z_boundary_matrix</code>         | $\mathbb{K}_9$ |
| <code>modular_S_commutator_norm</code> | 0.0            |
| <code>modular_T_commutator_norm</code> | 0.0            |
| <code>modular_invariant</code>         | True           |
| <code>integral_spin_closure</code>     | True           |
| <code>all_checks_passed</code>         | True           |

Thus the implemented boundary block satisfies

$$\|M_{\text{code}} S_{\partial}^{\text{code}} - S_{\partial}^{\text{code}} M_{\text{code}}\|_F = 0, \quad \|M_{\text{code}} T_{\partial}^{\text{code}} - T_{\partial}^{\text{code}} M_{\text{code}}\|_F = 0, \quad (63)$$

which is the finite visible-block realization of the modular closure condition used in the full SHBT boundary partition function.

## 4 Boundary Entropy Densities and Self-Resolution

The modular data of Section 3 also determine how the finite visible register is loaded. The relevant coordinate lattice is the  $3 \times 3$  visible block

$$\mathcal{C} = \{0, 1, 2\} \times \{0, 1, 2\}, \quad c = (i, j) \in \mathcal{C}. \quad (64)$$

The two axes of this lattice are the displayed  $SU(2)_{26}$  charge slice and the displayed  $SU(3)_8$  low-weight slice. We write

$$q_i, \ell_j \in (22, 23, 26), \quad w_i, w_j \in ((0, 0), (1, 0), (0, 1)), \quad (65)$$

where  $q_i$  and  $\ell_j$  are the row and column labels of the same visible  $SU(2)$  block. The raw boundary loading assigned to  $c = (i, j)$  is the product of the modular overlap probabilities in the two visible factors,

$$\tilde{\rho}_B(i, j) = \left| S_{q_i, \ell_j}^{SU(2)_{26}} \right|^2 \left| S_{w_i, w_j}^{SU(3)_8} \right|^2. \quad (66)$$

This is positive on the benchmark visible block, so the finite normalization constant

$$Z_B^{\text{load}} = \sum_{(a, b) \in \mathcal{C}} \tilde{\rho}_B(a, b) \quad (67)$$

is nonzero. The normalized loading density is therefore

$$\rho_B(i, j) = \frac{\tilde{\rho}_B(i, j)}{\sum_{(a, b) \in \mathcal{C}} \tilde{\rho}_B(a, b)}, \quad \sum_{(i, j) \in \mathcal{C}} \rho_B(i, j) = 1. \quad (68)$$

The associated entanglement density is the normalized Shannon contribution of each loaded coordinate,

$$\rho_E(i, j) = \frac{-\rho_B(i, j) \ln \rho_B(i, j)}{\sum_{(a, b) \in \mathcal{C}} -\rho_B(a, b) \ln \rho_B(a, b)}, \quad \sum_{(i, j) \in \mathcal{C}} \rho_E(i, j) = 1. \quad (69)$$

The convention  $0 \ln 0 = 0$  applies, although no zero entry occurs in the benchmark block.

The dominant loading sequence is the ordering of coordinates by decreasing normalized loading density,

$$\sigma = (c_1, \dots, c_9), \quad \rho_B(c_1) \geq \rho_B(c_2) \geq \dots \geq \rho_B(c_9). \quad (70)$$

Equal-density ties are physically degenerate and are resolved in the stable implementation order. This ordering is the audit field `dominant_loading_sequence`; it must agree with the coordinate order used by `sequence_bit_loading`. Equations (66)–(70) are implemented by the following audit helpers:

- `shbt_simulator.StaticBoundary.build_loading_density`,
- `shbt_simulator.StaticBoundary.build_entanglement_density`, and
- `shbt_simulator.StaticBoundary.build_dominant_sequence`.

Entropy self-resolution promotes the static loading order into an iterative finite-register update. Let  $U_n \subset \mathcal{C}$  be the unresolved set at step  $n$ , with  $c_n \in U_n$  the next coordinate selected by  $\sigma$ . Before the update, the cumulative resolved entropy state is

$$\Omega_{n-1}(i, j) = \sum_{m < n} \delta_{c_m, (i, j)} \frac{\Delta S_m}{N}. \quad (71)$$

The resolved state feeds back into each unresolved coordinate through the Manhattan-distance kernel

$$F_{n-1}(c) = \sum_{(i, j) \in \mathcal{C}} \frac{\Omega_{n-1}(i, j)}{1 + |i - c_i| + |j - c_j|}, \quad c = (c_i, c_j). \quad (72)$$

For feedback coupling  $\lambda \geq 0$ , the loading weight is dressed as

$$w_B(c) = \rho_B(c) (1 + \lambda F_{n-1}(c)). \quad (73)$$

The entropy channel is dressed by the same feedback kernel,

$$w_E(c) = \rho_E(c) (1 + \lambda F_{n-1}(c)). \quad (74)$$

With remaining bit and entropy budgets

$$R_{n-1}^B = N - \sum_{m < n} \Delta B_m, \quad R_{n-1}^E = N - \sum_{m < n} \Delta S_m, \quad (75)$$

the self-resolution update is

$$\Delta B_n = R_{n-1}^B \frac{w_B(c_n)}{\sum_{c \in U_n} w_B(c)}, \quad (76)$$

$$\Delta S_n = R_{n-1}^E \frac{w_E(c_n)}{\sum_{c \in U_n} w_E(c)}, \quad (77)$$

$$\Delta T_n = \frac{\Delta S_n}{N}. \quad (77)$$

Thus  $\Delta T_n$  is not an independent background clock tick. It is the dimensionless entropy residue left by resolving the next boundary coordinate. This update is the mathematical contract of `shbt_simulator.StaticBoundary.entropy_self_resolution()`, whose audit record stores `feedback_signal`, `delta_B`, `delta_S`, `delta_T`, and the updated unresolved set.

The dimensional projection is driven by the cumulative temporal residue. At step  $n$  the effective projection dimension is

$$D_n = 26 - 22 \sum_{m \leq n} \Delta T_m, \quad \Delta D_n = D_{n-1} - D_n = 22 \Delta T_n. \quad (78)$$

Since  $U_9$  contains only the final unresolved coordinate, the last update spends the full remaining entropy budget. By induction on Eq. (76),

$$\sum_{n=1}^9 \Delta S_n = N, \quad \sum_{n=1}^9 \Delta T_n = 1. \quad (79)$$

Consequently the benchmark projection starts at  $D_0 = 26$  and terminates at

$$D_9 = 26 - 22 = 4. \quad (80)$$

This proves the audit predicate `projection_dimension_26_to_4`: it holds precisely when the self-resolution run exhausts the normalized entropy budget and reports terminal dimension 4.

The same normalization gives the temporal kernel used for observer-frame cosmology. For a sampled expansion rate

$$H(t) = \frac{\dot{a}(t)}{a(t)}, \quad (81)$$

the total entropy-production rate is postulated to saturate the finite register as

$$\dot{S}_{\text{total}} = N H(t). \quad (82)$$

The local entanglement contribution is distributed by the normalized density,

$$\dot{S}_{ij} = \rho_E(i, j) \dot{S}_{\text{total}}. \quad (83)$$

Therefore

$$\dot{T} = \frac{1}{N} \sum_{(i,j) \in \mathcal{C}} \dot{S}_{ij} = \frac{\dot{S}_{\text{total}}}{N} \sum_{(i,j) \in \mathcal{C}} \rho_E(i, j) = H(t). \quad (84)$$

This is the perception identity: apparent temporal flow is the entropy flow per finite boundary bit. In the implementation it is audited by `shbt_simulator.StaticBoundary.derive_temporal_increment()`, with fields `local_entanglement_entropy_gradient`, `dot_S_total`, `dot_T`, and `perception_identity_holds`.

The full Section 4 implementation contract is therefore

| Mathematical object                    | Rust/Python simulator object              |
|----------------------------------------|-------------------------------------------|
| $\rho_B$                               | <code>loading_density</code>              |
| $\rho_E$                               | <code>entanglement_density</code>         |
| $\sigma$                               | <code>dominant_loading_sequence</code>    |
| $F_{n-1}$                              | <code>feedback_signal</code>              |
| $(\Delta B_n, \Delta S_n, \Delta T_n)$ | <code>entropy_self_resolution()</code>    |
| $D_9 = 4$                              | <code>projection_dimension_26_to_4</code> |
| $\dot{T} = H(t)$                       | <code>derive_temporal_increment()</code>  |

The boundary entropy densities are therefore not phenomenological weights added after modular closure. They are the normalized modular-overlap probabilities of the finite visible block, and their self-resolution produces both the  $26 \rightarrow 4$  dimensional projection and the local temporal kernel.

## 5 Holographic RG Flow: Boundary Entropy to Bulk Metric

The self-resolution cascade of Section 4 gives a finite entropy state on the visible boundary block. The holographic RG projector turns that state into a static bulk metric slice. In this section  $\tau \in [0, 1]$  denotes the normalized entropy-resolution parameter, not an independent external time coordinate. At the discrete audit steps,

$$\tau_n = \sum_{m \leq n} \Delta T_m = \frac{1}{N} \sum_{m \leq n} \Delta S_m, \quad \tau_0 = 0, \quad \tau_9 = 1. \quad (85)$$

Thus the RG parameter is the same finite-register entropy residue that drives the  $26 \rightarrow 4$  projection.

The prime lattice basis used by the metric projector is the five-point arithmetic skeleton

$$(p_0, p_1, p_2, p_3, p_4) = (2, 3, 5, 7, 11). \quad (86)$$

Its logarithms are the radial RG coordinates,

$$x_r = (\ln p_r), \quad r = 0, \dots, 4, \quad (87)$$

so nearest-neighbor finite differences on this lattice are naturally divided by  $\ln(p_{s+1}/p_s) = x_{s+1} - x_s$ . The role of the prime basis is not to introduce a new spectrum of particles. It is a fixed, finite coordinate frame in which the ordered boundary entropy state can be compressed into four Euler intervals and then lifted to a symmetric metric tensor.

Let

$$\sigma = (c_1, \dots, c_9) \quad (88)$$

be the dominant loading sequence of Eq. (70). The partially resolved entanglement state at RG parameter  $\tau$  is the interpolation of the discrete self-resolution records,

$$\Omega_\tau(c) = \sum_{n=1}^9 \theta_n(\tau) \delta_{c_n, c} \Delta T_n, \quad (89)$$

where

$$\theta_n(\tau) = \begin{cases} 0, & \tau < \tau_{n-1}, \\ \frac{\tau - \tau_{n-1}}{\tau_n - \tau_{n-1}}, & \tau_{n-1} \leq \tau \leq \tau_n, \\ 1, & \tau > \tau_n. \end{cases} \quad (90)$$

Consequently  $\sum_{c \in \mathcal{C}} \Omega_\tau(c) = \tau$ . The metric projector uses both the static entanglement density and the resolved entropy state. Along the ordered sequence define

$$E_m(\tau) = \rho_E(c_m) + \Omega_\tau(c_m), \quad m = 1, \dots, 9. \quad (91)$$

The five-component load vector is obtained by folding this ordered sequence onto the prime lattice by residue class,

$$\begin{aligned} \hat{\ell}_r(\tau) &= \sum_{m=1}^9 E_m(\tau) \mathbf{1}_{\{m-1 \equiv r \pmod{5}\}}, \quad r = 0, \dots, 4, \\ \ell_r(\tau) &= \frac{\hat{\ell}_r(\tau)}{\sum_{q=0}^4 \hat{\ell}_q(\tau)}. \end{aligned} \quad (92)$$

This gives

$$\ell_r(\tau) > 0, \quad \sum_{r=0}^4 \ell_r(\tau) = 1. \quad (93)$$

The dependence on  $\sigma$  is essential: the same multiset of boundary entropies would give a different prime-lattice load only if the modular loading order were changed. Equal-density ties remain the stable implementation ties already fixed in Section 4.

The Euler flux is the logarithmic finite difference of the load vector across adjacent prime sites. For  $s = 0, 1, 2, 3$ ,

$$\Phi_s(\tau) = \frac{\ell_{s+1}(\tau) - \ell_s(\tau)}{\ln(p_{s+1}/p_s)}. \quad (94)$$

Thus  $\Phi_s$  is the discrete RG gradient  $\Delta\ell/\Delta\ln p$  on the finite prime lattice. The local mean loading and arithmetic hardness factor are

$$\bar{\ell}_s(\tau) = \frac{\ell_s(\tau) + \ell_{s+1}(\tau)}{2}, \quad h_s = \frac{1}{p_s} + \frac{1}{p_{s+1}}. \quad (95)$$

The closure-weighted contribution of the  $s$ th Euler interval is

$$W_s(\tau) = \lambda_{\text{closure}} (|\Phi_s(\tau)| + \bar{\ell}_s(\tau) + h_s), \quad \lambda_{\text{closure}} > 0. \quad (96)$$

Every term in Eq. (96) is nonnegative, so the outer-product part of the metric projection is positive semidefinite. The absolute flux measures local entropy shear, the mean load measures local occupancy, and  $h_s$  prevents a completely flat load vector from erasing the arithmetic interval.

The four execution vectors embed the four Euler intervals into the static metric block. Let  $e_s^\mu = \delta_s^\mu$  for  $\mu, s = 0, 1, 2, 3$ , and read all load-vector indices modulo 5. The implemented execution vector is

$$v_s^\mu(\tau) = e_s^\mu + (\ell_s + \delta_{s0}, \ell_{s+1} + \delta_{s1}, \ell_{(s+2) \pmod 5} + \delta_{s2}, |\Phi_s| + \ell_{(s+3) \pmod 5} + \delta_{s3})^\mu. \quad (97)$$

The Kronecker terms mark the execution channel, while the cyclic load entries couple each interval to the neighboring prime sites. The fourth component also contains  $|\Phi_s|$ , so entropy shear enters the block as a metric direction rather than only as a scalar weight.

The unstabilized static metric is the weighted Gram matrix of these execution vectors, plus a positive diagonal closure floor,

$$\tilde{g}_{\mu\nu}(\tau) = \sum_{s=0}^3 W_s(\tau) v_{s\mu}(\tau) v_{s\nu}(\tau) + \text{diag}(d_0, d_1, d_2, d_3)_{\mu\nu}. \quad (98)$$

The diagonal entries used by the audit are

$$d_s(\tau) = \lambda_{\text{closure}} + W_s(\tau) + \ell_s(\tau) + \ell_{s+1}(\tau), \quad s = 0, 1, 2, 3. \quad (99)$$

Since  $\lambda_{\text{closure}} > 0$  and the remaining terms are nonnegative, Eq. (99) gives a strictly positive diagonal regularizer. The tensor in Eq. (98) is therefore already a finite, positive candidate for the static block; the final step enforces exact symmetry and trace normalization in the same way for every slice.

Define

$$A_{\mu\nu}(\tau) = \text{Sym}(\tilde{g}_{\mu\nu}(\tau) + \epsilon_{\text{eig}}\delta_{\mu\nu}), \quad \text{Sym}(B)_{\mu\nu} = \frac{B_{\mu\nu} + B_{\nu\mu}}{2}. \quad (100)$$

The eigenvalue lift is chosen as

$$\epsilon_{\text{eig}} = \max(0, \epsilon_{\text{pos}} - \lambda_{\min}(\text{Sym}(\tilde{g}))), \quad \epsilon_{\text{pos}} > 0. \quad (101)$$

In exact arithmetic the positive diagonal floor usually makes the lift unnecessary, but the audit keeps Eq. (101) so that numerical roundoff cannot produce a nonpositive metric slice. The physical static metric is

$$g_{\mu\nu}(\tau) = \frac{A_{\mu\nu}(\tau)}{\text{Tr } A(\tau)} = \frac{\text{Sym}(\tilde{g}_{\mu\nu}(\tau) + \epsilon_{\text{eig}}\delta_{\mu\nu})}{\text{Tr}[\text{Sym}(\tilde{g}(\tau) + \epsilon_{\text{eig}}\mathbf{1}_4)]}. \quad (102)$$

It obeys the metric-slice checks

$$g_{\mu\nu} = g_{\nu\mu}, \quad \text{Tr } g = 1, \quad \lambda_{\min}(g) > 0. \quad (103)$$

The trace condition fixes the finite-register scale of the slice; it does not remove physical anisotropy, because anisotropy is carried by the normalized eigenvalue ratios and off-diagonal components inherited from the execution vectors.

The spatial bulk metric is the projection of the static  $4 \times 4$  block onto the three rendered spatial directions. With

$$P_a{}^\mu = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad a = 1, 2, 3, \quad \mu = 0, 1, 2, 3, \quad (104)$$

the induced spatial metric is

$$g_{ab}^{\text{bulk}}(\tau) = P_a{}^\mu g_{\mu\nu}^{\text{static}}(\tau) P_b{}^\nu. \quad (105)$$

Equivalently, the spatial block is the lower-right  $3 \times 3$  principal block of the stabilized static metric in this benchmark projector. Other projectors could rotate the rendered spatial frame, but the audit uses the fixed matrix in Eq. (104) so that the benchmark is deterministic.

Combining the construction gives the complete holographic RG map

$$\boxed{\nabla_{\mathcal{C}} S_{\partial} \mapsto \rho_E(i, j) \mapsto \ell_r(\tau) \mapsto \Phi_s(\tau), W_s(\tau), v_s^\mu(\tau) \mapsto g_{\mu\nu}(\tau) \mapsto g_{ab}^{\text{bulk}}(\tau)}. \quad (106)$$

The shortened form is

$$\nabla_{\mathcal{C}} S_{\partial} \longrightarrow \rho_E(i, j) \longrightarrow \ell_r(\tau) \longrightarrow g_{\mu\nu}(\tau). \quad (107)$$

Every arrow is fixed by the previous sections: the boundary entropy gradient comes from modular-overlap loading,  $\rho_E$  is the normalized entanglement density,  $\ell_r$  is the prime-lattice compression of the ordered resolved state, and  $g_{\mu\nu}$  is the stabilized trace-one Gram projection. No independent bulk equation of motion is inserted between the boundary entropy state and the static metric slice.

The implementation contract for this section is

| Mathematical object                               | Rust/Python simulator object                      |
|---------------------------------------------------|---------------------------------------------------|
| RG projection controller                          | <code>shbt_simulator.HolographicProjection</code> |
| $\rho_E, \sigma, \Omega_\tau \mapsto \ell_r$      | <code>derive_load_vector()</code>                 |
| $\ell_r \mapsto \Phi_s, W_s, v_s^\mu, g_{\mu\nu}$ | <code>metric_from_load_vector()</code>            |
| $P_a^\mu g_{\mu\nu} P_b^\nu$                      | <code>project_static_block_to_bulk()</code>       |
| Symmetry, trace, positivity, spatial block        | <code>run_full_audit().projection_report</code>   |
| Metric slice record                               | <code>shbt_simulator.BulkMetricSlice</code>       |

The class `shbt_simulator.HolographicProjection` receives the finite entropy cascade from `shbt_simulator.StaticBoundary.entropy_self_resolution()`. Its `derive_load_vector()` method constructs Eq. (92); `metric_from_load_vector()` constructs Eqs. (94)–(102); `project_static_block_to_bulk()` applies Eq. (105); and `shbt_simulator.ShbtSimulator().run_full_audit().projection_report` checks Eq. (103), the  $3 \times 3$  shape of `spatial_metric`, agreement with the dominant loading sequence, and the absence of negative eigenvalues after stabilization. The benchmark therefore exports a list of `shbt_simulator.BulkMetricSlice` records whose fields include `load_vector`, `euler_flux`, `execution_vectors`, `unstabilized_metric`, `metric_components`, `spatial_metric`, and `eigenvalues`.

## 6 Topological Baryogenesis and Anti-Baryon De-Rendering

The baryogenesis sector uses the completed boundary branch to separate two questions that are usually entangled. The first is the size of the visible baryon excess. The second is the fate of the anti-baryon stress-energy that would ordinarily accompany visible anti-baryon charge. SHBT answers the first question with a topological CP invariant and a modular restoration scale, and it answers the second with an active de-rendering map that strips visible color and electromagnetic charge while preserving the passive gravitational source. The mechanism is therefore not a dynamical erasure of stress-energy. It is a projection of active render channels into the dark completion sector, constrained by the same framing closure that fixes the branch.

The visible electroweak conversion factor is the standard sphaleron coefficient for the rendered Standard Model channel,

$$C_{\text{sph}} = \frac{28}{79}. \quad (108)$$

This coefficient converts the topological lepton-sector seed into the baryon number observed in the active visible register. In SHBT the CP-odd seed is not introduced as a free phase. It is the level-locked topological Jarlskog invariant

$$J_{CP}^{\text{topo}} = \frac{1}{K} \left( \frac{k_q}{k_\ell + 2} \right) \sqrt{1 - \kappa_{D_5}^2} \sin \left( \frac{2\pi k_q}{k_\ell} \right). \quad (109)$$

The factors have distinct origins. The parent completion contributes the dilution  $1/K$ ; the visible level mismatch contributes  $k_q/(k_\ell + 2)$ , with the  $+2$  the  $SU(2)$  dual-Coxeter shift; the dark completion contributes the admissible overlap factor  $\sqrt{1 - \kappa_{D_5}^2}$ ; and the modular phase difference

between the quark and lepton clocks contributes  $\sin(2\pi k_q/k_\ell)$ . Thus CP violation is a residue of completed modular transport rather than an independently tunable angle.

The heavy restoration scale is fixed by the same branch arithmetic. Define the rank projection factor by

$$\Pi_{\text{rank}} = \sqrt{\frac{(h_{SO(10)}^\vee \dim SO(10))/12}{(h_{SU(3)}^\vee \dim SU(3))/12}} \sqrt{\frac{k_q + h_{SU(3)}^\vee}{K + h_{SO(10)}^\vee}}. \quad (110)$$

The residual matching of the 126 completion channel is written  $\delta\Pi_{126}^{\text{match}}$ . The modular restoration mass is then

$$M_N = M_{\text{GUT}} \exp\left[-\left(I_\ell^\star \Pi_{\text{rank}} + I_q^\star \delta\Pi_{126}^{\text{match}}\right)\right]. \quad (111)$$

For the canonical branch  $I_\ell^\star = 6$  and  $I_q^\star = 13$ , so the exponent is a structural suppression determined by the completed embedding. It is not adjusted after the baryon asymmetry is computed.

The SHBT baryogenesis identity is therefore

$$\eta_B = C_{\text{sph}} J_{CP}^{\text{topo}} \frac{M_N}{M_P}. \quad (112)$$

The four ingredients in Eq. (112) are fixed by the electroweak conversion coefficient, the modular CP residue, the restoration scale, and the Planck normalization. The benchmark implementation evaluates this identity as `eta_b=6.449923359416e-10`.

The visible asymmetry becomes a complete accounting statement only after the anti-baryon channel is specified. Let  $\mathcal{H}_B^{\text{vis}}$  denote the active visible anti-baryon Hilbert space. For a visible anti-baryon density matrix  $\rho$ , define the active render cost

$$\mathcal{C}_{\bar{B}}(\rho) = \gamma_3 \text{Tr}(\rho Q_{SU(3)}^2) + \gamma_{\text{EM}} \text{Tr}(\rho Q_{\text{EM}}^2) + \gamma_{\text{fr}} \Delta_{\text{fr}}^2, \quad \gamma_3, \gamma_{\text{EM}}, \gamma_{\text{fr}} > 0. \quad (113)$$

This is an active rendering functional, not a passive mass functional. It counts the cost of keeping anti-baryons visible as colored and electrically charged excitations, plus any residual framing mismatch. On the canonical branch  $\Delta_{\text{fr}} = 0$ , so the only nonzero active cost comes from visible gauge charge.

The de-rendering operator is the completion-sector projection

$$\mathcal{D}_{\bar{B}} : \mathcal{H}_B^{\text{vis}} \longrightarrow \mathcal{H}_{\text{dark}}^{B-L}, \quad \rho_{\bar{B}}^{(n+1)} = \frac{\mathcal{D}_{\bar{B}} \rho_{\bar{B}}^{(n)} \mathcal{D}_{\bar{B}}^\dagger}{\text{Tr}(\mathcal{D}_{\bar{B}} \rho_{\bar{B}}^{(n)} \mathcal{D}_{\bar{B}}^\dagger)}. \quad (114)$$

It is defined by the active charge-stripping constraints

$$\mathcal{D}_{\bar{B}}^\dagger Q_{SU(3)} \mathcal{D}_{\bar{B}} = 0, \quad \mathcal{D}_{\bar{B}}^\dagger Q_{\text{EM}} \mathcal{D}_{\bar{B}} = 0. \quad (115)$$

The dark target stores the conjugate  $B - L$  residue, so the completed boundary state remains neutral even though the active visible anti-baryon gauge channels are no longer rendered.

Stress-energy preservation is imposed on the passive source, not on the active charge cost. The defining equality is

$$\boxed{\text{Tr}\left(\rho^{(n)}T_{\mu\nu}^{\bar{B},\text{passive}}\right) = \text{Tr}\left(\rho^{(n+1)}T_{\mu\nu}^{\text{dark,passive}}\right)}. \quad (116)$$

Thus the visible register can lose active anti-baryon color and electromagnetic support without violating the gravitational accounting used by the topological Einstein equation. Equivalently, the de-rendered anti-baryon contributes as a dark passive source rather than as an actively rendered visible particle.

The anti-baryon loop closes at the fixed point

$$\boxed{\rho_{\bar{B}}^* = \frac{\mathcal{D}_{\bar{B}}\rho_{\bar{B}}^*\mathcal{D}_{\bar{B}}^\dagger}{\text{Tr}\left(\mathcal{D}_{\bar{B}}\rho_{\bar{B}}^*\mathcal{D}_{\bar{B}}^\dagger\right)}, \quad \mathcal{C}_{\bar{B}}(\rho_{\bar{B}}^*) = 0, \quad \delta T_{\mu\nu}^{\text{passive}} = 0.} \quad (117)$$

At this point the anti-baryon is absent from the active visible gauge-rendered channels, while its passive stress-energy has been moved into the dark completion sector. The observed baryon asymmetry is therefore a render-channel asymmetry, not an unbalanced disappearance of energy-momentum.

The implementation contract for this section is

| Mathematical object                                     | Rust/Python simulator object                        |
|---------------------------------------------------------|-----------------------------------------------------|
| Baryogenesis identity record                            | <code>shbt_simulator.BaryogenesisIdentity</code>    |
| $C_{\text{sph}}, J_{CP}^{\text{topo}}, M_N/M_P, \eta_B$ | <code>run_full_audit().baryogenesis_identity</code> |
| $\mathcal{C}_{\bar{B}}$                                 | <code>cpu_cycle_weight()</code>                     |
| $\mathcal{D}_{\bar{B}}$ charge stripping                | <code>derender_antibaryon_charges()</code>          |
| Field A versus Field B comparison                       | <code>run_full_audit().benchmark_delta</code>       |
| Passive source equality                                 | <code>report.stress_energy_preserved</code>         |

The simulator record `shbt_simulator.BaryogenesisIdentity` stores `sphaleron_coefficient`, `jarlskog_topological`, `Pi_rank`, `deltaPi_126_match`, `structural_exponent`, `modular_restoration_scale_gev`, `heavy_neutrino_to_planck_ratio`, and `eta_b`. The `shbt_simulator.ShbtSimulator().run_full_audit().baryogenesis_identity` record stores the numerical fields appearing in Eqs. (108)–(112). The method `shbt_simulator.BaryogenesisOptimizer.cpu_cycle_weight()` evaluates the active cost in Eq. (113); `shbt_simulator.BaryogenesisOptimizer.derender_antibaryon_charges()` implements the stripping constraints in Eq. (115); and `report.stress_energy_preserved` reports the passive equality check in Eq. (116). Finally, the `shbt_simulator.ShbtSimulator().run_full_audit().benchmark_delta` record compares the actively rendered Field A calculation with the SHBT-optimized Field B calculation, reporting the changes in `cpu_cycle_weight`, `operation_count`, `memory_bytes`, `elapsed_s`, and the Boolean `stress_energy_preserved`.

## 7 Causal Point and History Crystallization

The Causal Point is the observer interface of SHBT. It is not an external classical system appended to the boundary theory, but a finite local address inside the completed register. Its job is to translate global holographic closure into the quantities an observer can actually sample: a local memory budget, a local valuation of hidden information, a past-light-cone loading flow, and a deterministic projection that crystallizes one history from a finite outcome ensemble. Thus observation is treated as an entropy-budgeted operation on the boundary register rather than as a primitive background process.

The global horizon follows directly from the saturated bit budget introduced in the Introduction. Since  $N = 3\pi/(L_P^2 \Lambda_{\text{holo}})$ , the horizon radius is

$$R_H = \sqrt{\frac{NL_P^2}{\pi}} = \sqrt{\frac{3}{\Lambda_{\text{holo}}}}. \quad (118)$$

For a Causal Point whose local sampling radius is  $R_{\text{local}}$ , define the horizon fraction and local area by

$$f_H = \frac{R_{\text{local}}}{R_H}, \quad A_{\text{local}} = 4\pi R_{\text{local}}^2, \quad 0 \leq f_H \leq 1. \quad (119)$$

The observer cannot address the full horizon register unless  $f_H = 1$ . The available local memory, hidden complement, and operational entropy limit are therefore

$$N_{\text{local}} = N f_H^2, \quad N_{\text{hidden}} = N - N_{\text{local}}, \quad N_{\text{limit}} = \min\left(N_{\text{local}}, \frac{A_{\text{local}}}{4L_P^2 \ln 2}\right). \quad (120)$$

The first entry counts addressable register bits. The second counts degrees of freedom outside the local addressable patch. The third enforces the stricter of the local register count and the local area-law bound, with the factor  $\ln 2$  converting the area entropy into bits.

The Causal Point's self-valuation is the scalar response of the local observer to residual framing and hidden-sector loading. Writing  $\Delta_{\text{frame}} \equiv \Delta_{\text{fr}}$  and

$$f_{\text{hidden}} = \frac{N_{\text{hidden}}}{N} = 1 - \frac{N_{\text{local}}}{N}, \quad w(\xi) = \frac{\xi_\ell + \xi_q + \xi_s}{3}, \quad (121)$$

the valuation field is

$$\Sigma = (1 + \Delta_{\text{frame}})(1 + w(\xi)f_{\text{hidden}}). \quad (122)$$

On the canonical branch  $\Delta_{\text{frame}} = 0$ , but the formula is written in a branch-independent form so that any residual framing mismatch would be visible as a multiplicative observer cost. The localized entropy gradient is the valuation density across the local sampling radius,

$$\nabla_{\text{obs}} \Sigma = \frac{\Sigma f_{\text{hidden}}}{R_{\text{local}}}. \quad (123)$$

The apparent observer acceleration is then

$$a_{\text{obs}} = c^2 \nabla_{\text{obs}} \Sigma. \quad (124)$$

This acceleration is not inserted as an external force law. It is the local inertial response to the hidden entropy gradient seen by an entropy-limited observer.

The same local data define the observer-frame Jacobian. The anisotropy coordinates  $(\xi_\ell, \xi_q, \xi_s)$  are the three directional entries stored in the local property packet, and the Causal Point projection matrix is

$$J^\mu{}_\nu = \begin{pmatrix} 1 + \Sigma f_{\text{hidden}} & -\Sigma \xi_\ell & -\Sigma \xi_q & -\Sigma \xi_s \\ 0 & 1 + \Sigma \xi_\ell & 0 & 0 \\ 0 & 0 & 1 + \Sigma \xi_q & 0 \\ 0 & 0 & 0 & 1 + \Sigma \xi_s \end{pmatrix}. \quad (125)$$

Applied to a bulk metric slice from Section 5, it gives the observer-frame metric

$$g_{\mu\nu}^{\text{obs}} = J_\mu{}^\alpha g_{\alpha\beta} J_\nu{}^\beta. \quad (126)$$

The off-diagonal entries in the first row encode the finite-register mixing between temporal sampling and the three anisotropic property channels, while the diagonal spatial entries encode direction-dependent local rescaling.

Past-light-cone flow is the observer-time image of a static boundary loading process. Let  $N_{\text{load}}(t)$  be the number of loaded bits along the sampled history. The loading fraction and effective expansion rate are

$$f_{\text{load}}(t) = \frac{N_{\text{load}}(t)}{N}, \quad H_{\text{eff}}(t) = H_\Lambda + \frac{1}{3} \frac{df_{\text{load}}}{dt}. \quad (127)$$

Here  $H_\Lambda$  is the horizon rate associated with the static  $\Lambda_{\text{holo}}$  sector. The second term is not a new external clock; it is the one-dimensional residue of finite boundary loading as reconstructed by the Causal Point. Along the observer's past light cone, redshift parameterizes the same process through

$$\frac{dt}{dz} = -\frac{1}{(1+z)H_{\text{eff}}(z)}, \quad \frac{d\chi}{dz} = \frac{1}{H_{\text{eff}}(z)}. \quad (128)$$

The first equation gives the lookback-time increment, and the second gives the comoving radial increment in the same light-cone units. A sampled redshift therefore selects a boundary coordinate by the ordered loading sequence,

$$n(z) = 1 + \lfloor (|\sigma| - 1) f_{\text{load}}(z) \rfloor, \quad c(z) = \sigma_{n(z)} \in \mathcal{C}. \quad (129)$$

This gives the bridge between the apparent cosmological flow and the finite boundary address on which a measurement can be attempted.

History crystallization begins only after such a boundary address has been selected. Let  $\mathcal{R}$  be the finite register visible to the Causal Point, let  $A \in \mathcal{R}$  be the selected address, and let

$$\Omega_A = \{\omega_0, \omega_1, \dots, \omega_{m-1}\}, \quad \sum_{i=0}^{m-1} p_i = 1, \quad p_i \geq 0, \quad (130)$$

be the finite outcome ensemble and its normalized local loading weights. The entropy costs of a retrieval event are

$$\begin{aligned} C_{\text{addr}} &= \log_2 |\mathcal{R}|, & C_{\text{ens}} &= \log_2 |\Omega_A|, \\ C_{\text{get}} &= \max(1, C_{\text{addr}} + C_{\text{ens}}, C_{\text{req}}), & R_{\text{entropy}} &= N_{\text{limit}} - C_{\text{get}}. \end{aligned} \quad (131)$$

The GET operation is admissible exactly when

$$R_{\text{entropy}} \geq 0. \quad (132)$$

If the residual is negative, the Causal Point has insufficient local entropy budget to crystallize the requested classical record.

When the budget condition holds, the deterministic collapse index is

$$\iota = \text{Hash}(\text{observable\_name}, A, f_H, N_{\text{local}}, m) \bmod m. \quad (133)$$

The corresponding rank-one projection operator is

$$\Pi_{A,\iota} = |A, \omega_\iota\rangle\langle A, \omega_\iota|. \quad (134)$$

For a sequence of admissible Causal Point retrievals, the history density matrix updates by

$$\rho_{\text{hist}}^{(n+1)} = \frac{\Pi_{A_n, \iota_n} \rho_{\text{hist}}^{(n)} \Pi_{A_n, \iota_n}}{\text{Tr}(\Pi_{A_n, \iota_n} \rho_{\text{hist}}^{(n)})}. \quad (135)$$

The full crystallized history along a sampled past light cone is therefore the ordered product

$$\Pi_{\text{history}}(z_1, \dots, z_m) = \Pi_{A_m, \iota_m} \cdots \Pi_{A_2, \iota_2} \Pi_{A_1, \iota_1}, \quad A_n = A(c(z_n)). \quad (136)$$

Thus a classical past is not retrieved from an unlimited archive. It is the finite ordered record that survives all local entropy-budget checks.

The pointer wavefunction associated with the selected outcome is the visible and dark completion packet

$$\Psi_\iota = (\text{amplitude} = p_\iota, \quad \text{c\_vis} = -p_\iota, \quad \text{c\_dark} = p_\iota). \quad (137)$$

The opposite signs of the visible and dark coefficients encode the same completion accounting used elsewhere in SHBT: the visible pointer is rendered as a local classical record while the complementary dark coefficient preserves the finite-register balance.

The implementation contract for this section is

| Mathematical object                   | Rust/Python simulator object                |
|---------------------------------------|---------------------------------------------|
| Causal observer interface             | <code>shbt_simulator.CausalPoint</code>     |
| Past-light-cone samples               | <code>build_past_light_cone()</code>        |
| Local anisotropy and packet data      | <code>compute_property_packets()</code>     |
| Entropy-budget admissibility          | <code>run_full_audit().memory_report</code> |
| History projection and pointer packet | <code>crystallize_history()</code>          |

The class `shbt_simulator.CausalPoint` stores the local horizon fraction, hidden-bit fraction, self-valuation, observer Jacobian, and past-light-cone samples. Its `build_past_light_cone()` method evaluates Eqs. (127)–(129); `compute_property_packets()` stores the anisotropy data entering Eqs. (122) and (125); `shbt_simulator.ShbtSimulator().run_full_audit().memory_report`

checks Eqs. (120) and (132); and `crystallize_history()` constructs Eqs. (133)–(137). The benchmark fields reported by this interface include `local_available_bits`, `hidden_bits`, `entropy_limit_bits`, `sigma`, `localized_entropy_gradient_per_m`, `gravitational_acceleration_m_per_s2`, `collapse_index`, `retrieval_cost_bits`, `entropy_budget_residual`, and `pointer_wavefunction`.

## 8 Numerical Verification and Performance

The preceding sections define SHBT by algebraic closure, entropy self-resolution, holographic projection, Causal Point admissibility, and anti-baryon de-rendering. This section records the corresponding finite audit outputs from the canonical Rust/Python simulator. The simulator entry point is `shbt_simulator.ShbtSimulator().run_full_audit()`, and the reproducibility command is `python examples/run_audit.py`. The older standalone `shbt_core.py` audit is deprecated and removed from the paper-level specification; where legacy values differ, the new simulator output is taken as canonical because the audit uses 512-bit precision and parallel computation. The verification is not a separate phenomenological fit: every entry is evaluated on the same canonical branch

$$\boxed{\text{shbt\_simulator.StaticBoundary.benchmark\_branch} = (26, 8, 312), \quad \Delta_{\text{fr}} = 0.} \quad (138)$$

The purpose of the numerical layer is therefore to check that the branch data, normalization conventions, projector algebra, observer memory accounting, and baryogenesis optimizer all close simultaneously.

### 8.1 Boundary closure audit

The `shbt_simulator.StaticBoundary` component tests the canonical branch before any bulk metric or observer-frame reconstruction is used. The first requirement is exact framing closure. The audit evaluates

$$\Delta_{\text{fr}}(312, 26, 8) = \max\left(\left\|\frac{312}{2 \cdot 26}\right\|_{\mathbb{Z}}, \left\|\frac{312}{3 \cdot 8}\right\|_{\mathbb{Z}}\right) = 0, \quad (139)$$

and then checks that the loading and entanglement densities used by the entropy flow are normalized. The modular tests verify that the visible and completed partition data commute with the constructed  $S_{\partial}$  and  $T_{\partial}$  kernels to simulator precision:

$$\|[M, S_{\partial}]\| = 0.0, \quad \|[M, T_{\partial}]\| = 0.0. \quad (140)$$

The full boundary audit is summarized in Table 1.

Together these checks establish the implementation form of the first closure equivalence:

$$[M, S_{\partial}] = [M, T_{\partial}] = 0 \iff \Delta_{\text{fr}} = 0 \implies \Pi_{\text{phys}} H_{\partial} \Pi_{\text{phys}} = 0. \quad (141)$$

The final two entries in Table 1 are especially important for reproducibility: `report.zero_energy_locked=True` confirms that the completed static boundary has no

| Audit quantity                  | Simulator accessor                                | Rust/Python output               |
|---------------------------------|---------------------------------------------------|----------------------------------|
| Branch levels                   | <code>report.branch</code>                        | (26, 8, 312)                     |
| Framing defect                  | <code>report.framing_defect</code>                | 0.0                              |
| $Z_{\partial}^{\text{code}}(i)$ | <code>evaluate_z_boundary()</code>                | 2.441381789163 $\times 10^{-24}$ |
| Loading density sum             | <code>build_loading_density()</code>              | 1.0                              |
| Entanglement density sum        | <code>build_entanglement_density()</code>         | 1.0                              |
| Dominant sequence length        | <code>build_dominant_sequence()</code>            | 9                                |
| Modular $S$ commutator          | <code>boundary_report.modular_S_commutator</code> | 0.0                              |
| Modular $T$ commutator          | <code>boundary_report.modular_T_commutator</code> | 0.0                              |
| Modular invariant               | <code>report.modular_invariant</code>             | true                             |
| Zero-energy lock                | <code>report.zero_energy_locked</code>            | true                             |
| Visible projection              | <code>report.projection_dimension_26_to_4</code>  | true                             |

Table 1: Static boundary verification results from the canonical Rust/Python simulator.

residual external time generator, while `report.projection_dimension_26_to_4=True` confirms that the visible lepton-sector projection used by the benchmark branch has the expected finite dimension.

## 8.2 Bulk metric projection audit

The `shbt_simulator.HolographicProjection` component starts from the normalized loading and entanglement densities and constructs the entropy cascade of metric slices. The benchmark run returns the full nine-slice cascade,

$$\text{projection\_report.slice\_count} = 9, \quad (142)$$

with a rank-three spatial projector at every sampled step. Each metric slice is then checked for the three algebraic properties required by the projection construction:

$$g_{ij} = g_{ji}, \quad \text{Tr}(g) = 1, \quad v^i g_{ij} v^j > 0 \quad \text{for } v \neq 0. \quad (143)$$

The resulting audit table is Table 2.

| Audit quantity         | Simulator accessor                                   | Rust/Python output |
|------------------------|------------------------------------------------------|--------------------|
| Entropy cascade length | <code>projection_report.slice_count</code>           | 9                  |
| Spatial projector rank | <code>projection_report.projector_rank</code>        | 3                  |
| Metric symmetry        | <code>projection_report.symmetric</code>             | true               |
| Trace normalization    | <code>projection_report.trace_normalized</code>      | true               |
| Positive definiteness  | <code>projection_report.positive_definiteness</code> | true               |

Table 2: Holographic projection verification results from the canonical Rust/Python simulator.

This confirms that the entropy flow does not merely produce arrays of numbers. It produces symmetric, trace-normalized, positive metric data with the intended rank-three spatial image, so the bulk slices satisfy the numerical version of the projection conditions stated in Section 5.

### 8.3 Causal Point audit

The `shbt_simulator.CausalPoint` component verifies that observer access is limited by a finite entropy budget. For a horizon fraction  $f_H$ , the local bit budget is checked against

$$\text{memory\_report.local\_available\_bits} \simeq N f_H^2, \quad \text{memory\_report.hidden\_bits} = N - \text{memory\_report.local\_available\_bits} \quad (144)$$

The retrieval interface is admissible only when the entropy limit is positive, and the past-light-cone builder is checked against the requested redshift grid:

$$\text{memory\_report.entropy\_limit\_bits} > 0, \quad \text{memory\_report.past\_light\_cone\_samples} = \text{redshift\_samples} \quad (145)$$

The audit is summarized in Table 3.

| Audit quantity       | Simulator accessor                                 | Rust/Python output                  |
|----------------------|----------------------------------------------------|-------------------------------------|
| Local available bits | <code>memory_report.local_available_bits</code>    | $2.535748254483999 \times 10^{122}$ |
| Hidden bits          | <code>memory_report.hidden_bits</code>             | $7.76249465658367 \times 10^{121}$  |
| Entropy limit        | <code>memory_report.entropy_limit_bits</code>      | $2.535748254483999 \times 10^{122}$ |
| Light-cone samples   | <code>memory_report.past_light_cone_samples</code> | 9                                   |
| Property packets     | <code>memory_report.property_packets</code>        | 9                                   |
| Memory admissibility | <code>report.memory_all_passed</code>              | true                                |

Table 3: Causal Point verification results from the canonical Rust/Python simulator.

These outputs check the observer side of the closure chain. The Causal Point does not access an unlimited record; it constructs exactly as many past-light cone entries as the redshift sampler asks for, constructs nine local property packets, and filters each attempted retrieval through the finite-memory inequality.

### 8.4 Baryogenesis and optimized field simulation

The baryogenesis benchmark compares a standard visible field simulation with the optimized SHBT render in which anti-baryon visible charge channels are de-rendered while their passive stress-energy contribution remains in the dark completion sector. The simulator reports the visible baryon asymmetry as

$$\eta_B = 6.449923359416 \times 10^{-10}, \quad (146)$$

with `report.stress_energy_preserved=True`. In the benchmark implementation the optimization is also a performance statement: fewer actively rendered field channels are evolved, while the passive gravitational accounting is unchanged. The standard and optimized simulations are compared in Table 4.

The physical comparison is shown in Table 5. The optimized simulation is not allowed to discard

| Simulation mode           | Active visible channels                  | CPU cycles | Operations                        | Memory |
|---------------------------|------------------------------------------|------------|-----------------------------------|--------|
| Standard field simulation | baryon and anti-baryon                   | 1.0        | 1.0                               | 1.0    |
| Optimized SHBT simulation | baryon rendered, anti-baryon de-rendered | 0.5        | 0.00153331474770.0058214747736093 |        |
| Reduction fraction        | removed active anti-baryon render        | 0.5        | 0.99846668525230.9941785252263907 |        |

Table 4: Normalized standard-versus-optimized field simulation cost from `benchmark_delta`.

energy: it removes anti-baryon charge from the actively rendered visible channel but preserves the passive source that enters the topological Einstein equation.

| Quantity                    | Standard simulation        | Optimized SHBT simulation                 |
|-----------------------------|----------------------------|-------------------------------------------|
| Visible baryon channel      | actively rendered          | actively rendered                         |
| Visible anti-baryon channel | actively rendered          | de-rendered                               |
| Passive stress-energy       | explicitly evolved         | preserved in dark completion              |
| Baryon asymmetry output     | dynamical baseline target  | $\eta_B = 6.449923359416 \times 10^{-10}$ |
| Stress-energy check         | baseline conservation test | true                                      |

Table 5: Physical accounting in the standard and optimized simulations.

The combined verification suite therefore checks both correctness and cost. On the correctness side, `shbt_simulator.ShbtSimulator().run_full_audit()` returns the same branch data with `report.modular_invariant=True`, `report.zero_energy_locked=True`, `report.memory_all_passed=True`, and `report.stress_energy_preserved=True`. On the performance side, the optimized render reduces CPU cycles by 50%, operation count by approximately 99.8467%, and memory use by approximately 99.4179% relative to the normalized standard field simulation, without violating the stress-energy preservation identity.

## 9 Unification and Conclusions

The preceding sections describe SHBT from four complementary directions: modular completion of the boundary theory, entropy self-resolution into bulk metric data, anti-baryon de-rendering with passive stress-energy preservation, and entropy-budgeted observer history. The unifying statement is that these are not independent mechanisms. They are different projections of the same finite closed boundary register. Once the completed partition function is modular invariant and the canonical branch arithmetic closes, the bulk tensor, the physical Hamiltonian constraint, the baryogenesis fixed point, and the Causal Point history rule all become consequences of the same closure data.

The core closure chain is

$$[M, S_\partial] = [M, T_\partial] = 0 \iff \Delta_{\text{fr}} = 0 \iff \mathcal{E}_{\mu\nu} = 0 \iff \Pi_{\text{phys}} H_{\text{total}} \Pi_{\text{phys}} = 0. \quad (147)$$

Equivalently,  $H_{\text{total}} = 0$  after restriction to the physical subspace. The first condition says that the torus partition function is a genuine modular invariant of the completed boundary CFT. The second says that the visible levels and parent completion level have no residual framing mismatch. The third says that the bulk closure tensor vanishes, so the SHBT topological Einstein equation is satisfied without adding an independent bulk correction. The fourth says that the static physical sector has no external time generator left over after projection. Thus the canonical branch is not merely a useful choice of numbers; it is the branch on which boundary modular covariance, visible-sector anomaly closure, bulk geometry, and the Hamiltonian constraint are the same statement viewed through different interfaces.

The geometric part of the construction is summarized by the holographic RG map

$$\boxed{\nabla_C S_\partial \longrightarrow \rho_E(i, j) \longrightarrow \ell_r(\tau) \longrightarrow g_{\mu\nu}(\tau).} \quad (148)$$

The boundary entropy gradient is first converted into the normalized entanglement density  $\rho_E(i, j)$  on the finite modular-overlap coordinate set. The ordered resolved state is then compressed into the prime-lattice load vector  $\ell_r(\tau)$ , and the stabilized Gram projection of this load vector gives the effective metric slice  $g_{\mu\nu}(\tau)$ . In this sense geometry is not a continuous background into which the boundary theory is inserted. It is the rendered image of finite boundary information after entropy gradients have been normalized, ordered, compressed, and projected.

The matter-asymmetry sector closes by the anti-baryon de-rendering loop

$$\boxed{\rho_B^* = \frac{\mathcal{D}_B \rho_B^* \mathcal{D}_B^\dagger}{\text{Tr}(\mathcal{D}_B \rho_B^* \mathcal{D}_B^\dagger)}, \quad \mathcal{C}_B(\rho_B^*) = 0, \quad \delta T_{\mu\nu}^{\text{passive}} = 0.} \quad (149)$$

At the fixed point, active anti-baryon color and electromagnetic render charge have been stripped from the visible channel, but the passive gravitational source has not been discarded. It is carried by the dark completion sector. The resulting baryon excess is therefore an asymmetry of rendered visible charge, not a violation of the completed stress-energy account. This is why the same branch that fixes  $\Delta_{\text{fr}} = 0$  also supports `stress_energy_preserved=True`: the de-rendering map is allowed only because it is a completion-preserving map.

The observer sector supplies the final link in the same chain. A Causal Point does not read a pre-existing classical history from an infinite archive. It crystallizes a finite ordered sequence of admissible projections,

$$\boxed{\Pi_{\text{history}} = \Pi_{A_m, \ell_m} \cdots \Pi_{A_2, \ell_2} \Pi_{A_1, \ell_1}.} \quad (150)$$

Each factor is admitted only if the local entropy budget remains nonnegative. The apparent classical past is therefore the record that survives the ordered application of finite-register GET operations. The observer is part of the cosmic code rather than an external reader of it: the address  $A_n$ , outcome index  $\ell_n$ , local hidden-state budget, and pointer packet are all internal boundary data.

The principal numerical predictions of the benchmark branch can be collected as

$$\boxed{\Lambda_{\text{holo}} \simeq 1.09 \times 10^{-52} \text{ m}^{-2}, \quad N \simeq 3.31 \times 10^{122}, \quad \eta_B = \mathcal{O}(10^{-10}), \quad g_{\mu\nu} = g_{\mu\nu}(\nabla_C S_\partial).} \quad (151)$$

The first number is the holographic dark-energy curvature scale used by the topological Einstein equation. The second is the finite boundary bit budget associated with that scale. The third is the order of the visible baryon asymmetry produced by the de-rendering loop, with the benchmark audit giving  $\text{eta\_b}=6.449923359416\text{e-}10$ . The fourth is the structural prediction: the effective metric is not fundamental input but a calculable output of the entropy-gradient pipeline.

Relative to standard holographic approaches, SHBT keeps the boundary-first logic but changes what is taken as primitive. It does not start from a large- $N$  gauge theory with an asymptotically anti-de Sitter bulk and then ask for a semiclassical interior. It starts from a completed static boundary CFT, requires exact modular closure, and lets finite entropy data render the bulk metric. Relative to entropic-gravity pictures, SHBT does not infer gravity from entropy alone; it constrains the entropy flow by the modular invariant pairing matrix, the canonical branch arithmetic, and the finite register size. Relative to purely constraint-based quantum gravity, it does not stop at the formal statement that the Hamiltonian vanishes on physical states; it ties that statement to visible-sector levels, dark completion, baryogenesis, and observer memory.

The most direct tests of the framework are therefore consistency tests across sectors. The branch data must give integer locks  $I_\ell^* = 6$  and  $I_q^* = 13$ , zero framing defect, a vanishing bulk closure tensor, and a finite physical-sector Hamiltonian constraint. The entropy pipeline must produce symmetric, trace-normalized, positive metric slices from the same boundary densities that define the loading sequence. The baryogenesis module must remove active visible anti-baryon charge while keeping passive stress-energy unchanged. The Causal Point module must crystallize only those histories whose retrieval cost fits inside the local entropy budget. Any failure of one of these checks would be a failure of the unified SHBT closure, not merely a failure of an auxiliary numerical routine.

Several limitations remain. The present paper establishes and audits one canonical benchmark branch rather than a classification theorem for all modular completions. The dark completion sector is fixed by closure requirements but still needs a more explicit microscopic construction. The entropy-to-metric map is deterministic in the benchmark implementation, yet a full comparison with precision cosmological data would require a larger phenomenological layer: survey observables, perturbations, noise models, and parameter-estimation pipelines. The baryogenesis mechanism also needs extension from the fixed render-channel identity to a complete account of early-universe thermal history as seen by finite boundary observers. These are not optional decorations; they are the natural next checks if SHBT is to move from a closed auditable framework to a quantitatively predictive research program.

The cosmic-code perspective is the simplest way to state the conclusion. A finite boundary register, when forced to close under completed modular kernels, contains more than a list of states. It contains the rules for which visible charges can be rendered, which passive sources must be preserved, which entropy gradients become geometry, and which observer histories can crystallize. In this view spacetime is the readable projection of a finite holographic code; matter asymmetry is a render-channel fixed point of that code; and observation is the entropy-budgeted act of selecting a consistent history from within the same code. SHBT is therefore static only at the level of the completed boundary. The dynamical world seen by an observer is the internally crystallized image of that closed finite boundary information.

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